PROJECT-BASED LEARNING

PHASE-I EVALUATION

Smart Water Management and Irrigation Optimization System Using C++

Team ID: T294

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T294
Semester: 3rd
Section: ML1, ML2, ML4
Domain: *SMART WATER MANAGEMENT*

Team Members

1. *Akanksha Katiyar – 2510378334*
2. *Pragya Anand – 2510011890*
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Mentor

Dr. Narayan Chaturvedi

Interactions completed:3

Problem Statement & Motivation

Problem Statement

Traditional irrigation may distribute water unevenly across farm zones, causing over-irrigation in some plots and under-irrigation in others. Water and electricity can also be wasted. The project aims to prioritize the zones that need irrigation most.

Motivation

- *Use soil moisture, temperature and rainfall probability.*
- *Prioritize zones with higher irrigation need.*
- *Prototype simulates pumps and sensors rather than controlling*

Target Users / Stakeholders

- *Farm zones and irrigation management scenarios.*
- *Users entering or simulating sensor readings.*
- *Future agricultural systems that could integrate hardware.*

Objectives

Project Objectives

1. *Develop a C++ application to identify irrigation needs of different farm zones.*
2. *Create data models for farm zones, sensors and irrigation details.*
3. *Process sensor readings using a queue.*
4. *Calculate irrigation priority from moisture, temperature and rainfall probability.*
5. *Use a priority queue to select the zone with the highest irrigation requirement.*
6. *Model the water-pipe network as a graph.*

Key Objective: *Distribute limited water according to the priority of farm zones.*

Proposed Solution

Proposed Approach

1. *Simulate or enter sensor values.*
2. *Store incoming readings in a queue.*
3. *Calculate irrigation priority for each zone.*
4. *Select the next zone using a priority queue.*
5. *Find an efficient water-delivery route using a graph and Dijkstra's algorithm.*

Key Features

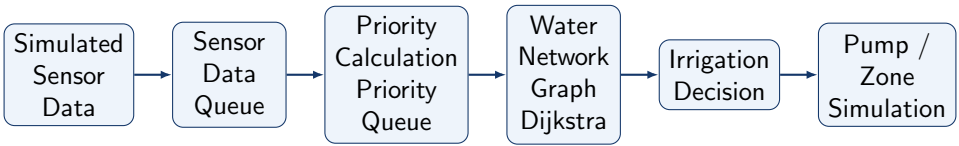
- *Sensor-data simulation*
- *Priority-based water allocation*
- *Graph-based water network*
- *Dijkstra shortest-path routing*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Queue, priority queue, graph, adjacency list, linked list/vector*	*Sensor processing, priority, network, history*
OOPs with C++	*Dijkstra's shortest-path algorithm; priority-based selection*	*Efficient water route and irrigation order*

Requirement: the project combines data structures, algorithms and C++ implementation in **one** irrigation simulation.

System Architecture / Workflow



Major Components

- *Farm-zone data model*
- *Sensor processing module*
- *Priority logic*
- *Water-network graph*

Data / Control Flow

- *Sensor values enter the queue for processing.*
- *Priority queue chooses the most needy irrigation zone.*
- *Graph + Dijkstra determines an efficient delivery route.*

Technology Stack & Methodology

Technology Stack

- Programming: *C++*
- Library: *C++ Standard Library*
- Data structures: *queue, priority queue, graph, adjacency list, linked list/vector*
- Algorithm: *Dijkstra's shortest path*
- Version Control: *GitHub*

Development Methodology

1. Requirement and problem analysis
2. System and data-structure design
3. C++ implementation
4. Scenario-based testing

Initial Progress & Team Contribution

Progress Achieved

- *Project proposal and motivation defined.*
- *Goals and milestones identified.*
- *Core data structures and Dijkstra algorithm selected.*

Individual Contribution

Member	Role	Contribution
Akanksha	*Lead*	*Team leadership and project coordination*
Pragya	*Team Member*	*Report formation*
Shivendra	*Team Member*	*Data collection*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- *Working C++ console-based smart irrigation prototype*
- *Priority-based allocation of limited water*
- *Efficient route calculation using Dijkstra*

References

1. *C++ reference.*
2. *Documentation for queue, priority queue, vector and unordered map*
3. *Standard references on graphs, heaps and Dijkstra's algorithm*
4. *FAO resources on efficient irrigation and agricultural water management*
5. *Course notes*

Thank You

Questions & Suggestions